quantum fluctuations cause different parts to approach at slightly different rates. When they finally slam together roughly a trillion years later, different locations on the branes will make contact at slightly different moments, rather as if two pieces of coarse sandpaper were being slapped together. The tiny deviations from a perfectly uniform impact yield tiny deviations from a perfectly uniform evolution across each brane. Since one of these branes is supposed to be our three-dimensional space, the deviations from uniformity are deviations we should be able to detect. Steinhardt, Turok, and their collaborators have argued that the inhomogeneities generate temperature deviations of the same form as those emerging from the inflationary framework, and hence, with today's data, the cyclic model offers an equally viable explanation of the observations.

However, the more refined data being gathered over the next decade may be able to distinguish between the two approaches. In the inflationary framework, not only are quantum fluctuations of the inflaton field stretched by the burst of exponential expansion, but tiny quantum ripples in the spatial fabric are also generated by the intense outward stretching. Since ripples in space are nothing but gravitational waves (as in our earlier discussion of LIGO), the inflationary framework predicts that gravitational waves were produced in the earliest moments of the universe.[8] These are often called *primordial gravitational waves,* to distinguish them from those generated more recently by violent astrophysical events. In the cyclic model, by contrast, the deviation from perfect uniformity is built up gently, over the course of an almost unfathomable length of time, as the branes spend a trillion years slowly heading toward their next splat. The absence of a brisk and vigorous change in the geometry of the branes, and in the geometry of space, means that spatial ripples are *not* generated, so the cyclic model predicts an absence of primordial gravitational waves. Thus, if primordial cosmological gravitational waves should be detected, it will be yet another triumph for the inflationary framework and will definitively rule out the cyclic approach.

It is unlikely that LIGO will be sensitive enough to detect inflation's predicted gravitational waves, but it is possible that they will be observed indirectly either by Planck or by another satellite experiment called the Cosmic Microwave Background Polarization experiment (CMBPol) that